WELCOME!

Online Training Course

Course 2: Foundations in Mobile Phone Data (MPD) for Policy: Modern Data Workflows

Course 2: Foundations in Mobile Phone Data (MPD) for Policy: Modern Data Workflows

Overview

Eight online training sessions (live webinars):

- Designing mobile phone data for policy pipelines
- Privacy-by-design
- Input data quality assurance and cleaning
- Continuity models
- PoIs/Home Locations / Usual Environment
- Calculating "foundational" / "core" aggregates
- Quality Assessment of Estimates for Decision-Making
- From Aggregates to Estimates - population inference from mobile phone data and standard data sources

Five self-paced topics (on OLC):

- Recap of fundamentals of mobile phone data, data pipelines, data products, data types
- "Introduction to the course and examples use cases: data pipeline and stages
- Skill sets for each stage. Objectives."
- Mobile Phone Data: datasets, fields, types (links with 1.5)
- Reference and Complementary data etc.
- Considerations for multi-operators data pipelines

Housekeeping information

- Delivery timeslot for online training sessions (live webinars):
  - Every Wednesday between 22 October and 10 December
  - From **12-1:30pm UTC** on **22 October**
  - From **1pm-2pm UTC** from **29 October to 10 December**

- Course 1 will only be available until end of October
- The slides and a session recording will be uploaded to the Open Learning Campus after delivery, for your reference
- Additional resources, quizzes and/or other follow-up will also be made available after sessions
- Try to keep Q&A until the end

Designing mobile phone data for policy Pipelines

Lesson 2.3.2

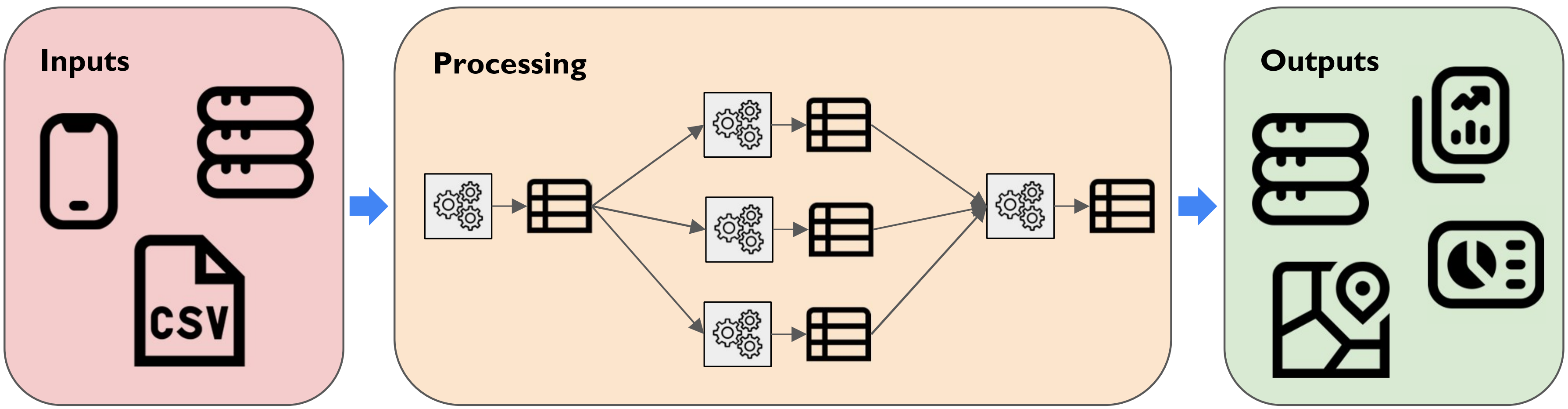
Introduction

This session will cover the design of data pipelines for MPD-for-policy use cases. We will cover:

- Types of input and output data
- Qualities that a good data pipeline should have
- The stages of a data processing pipeline
- Considerations that influence the design of a data pipeline

What is a data pipeline?

A system that transforms, processes and moves data from one or more sources to one or more destinations.



Icons: Flaticon.com

Learning objectives

- Identify internal and external **stakeholders** regarding the development of a data pipeline for a MPD-for-policy initiative
- Describe the types of **outputs** that the data pipeline will produce
- Describe how to find **where** required **input data** is stored
- Explain why MPD data should be **standardised** for further processing
- Describe the **qualities** of a “good” MPD for Policy pipeline
- Describe the **considerations** that factor into the **requirements** of the data pipeline
- Understand factors that determine **storage and processing infrastructure** requirements

Contents

Recap of pre-reading material

Aligning and managing stakeholders

- Providers:
  - Owners and controllers of input datasets
  - E.g. staff within MNO or NSI
  - Source of information about location/size/quality/structure of data
- Maintainers:
  - Responsible for building and maintaining the pipeline
  - May include staff in indirect supporting roles
  - Need to be aligned on pipeline aims and roles
- Consumers:
  - Users of pipeline outputs
  - Includes direct users, but also indirect users such as policy makers and affected population
  - Consumers' needs determine the content, format, timing and frequency of outputs

Types of outputs

- Intermediate datasets
- Indicators
- Diagnostics
- Maps and visualisations
- Reports
- Dashboard

Format, frequency and dissemination method of outputs depend on use case - strongly influences pipeline design.

Navigating MNO data infrastructure

Where are the source data?

Before building a pipeline, we need to find the required data.

Mobility analysis may require combining MNO data managed by different teams, even stored in different databases:

- CDR data
  - Billing / BI team?
- Other positioning data - similar structure to CDR but recorded for different purposes
  - E.g. network operations?
- Cell positioning/coverage data
  - Radio networks team
- Client portfolio data (handset type, customer segmentation, …)
  - Commercial / BI team?

Where are the source data?

Need to identify:

- Where are the different datasets?
  - What database? How are they accessed?
- Who is responsible for each?
  - Which team? Identify a point of contact
- What quality checks are applied/monitored?
  - Missing data? Invalid IDs?
- How frequently/quickly is each dataset updated?
  - Daily? Continuously? How long before new data are available?
- How to combine datasets
  - E.g. what fields are used to join CDR events to cell locations?

Qualities of a good data pipeline

Qualities of a good data pipeline

- Data standardisation
- Quality assurance
- Recurrent and reproducible processing
- Maintainability
- Performance and efficiency
- Security and data governance

Qualities of a good data pipeline

- Data standardisation
- Quality assurance
- Recurrent and reproducible processing
- Maintainability
- Performance and efficiency
- Security and data governance

Data standardisation

What is data standardisation?

- **Consistent structure** and format (column names, data types, text value formats)
- **Documented** for convenient reference and clarity
- Applies to data at **all stages of a pipeline** (inputs, outputs and intermediate tables)
- Where possible, **enforced using constraints** (not-null, unique, foreign key, …)
- Outputs align with accepted **standards** where relevant

Data standardisation

Why is it important?

- **Accuracy -** errors and data quality issues are prevented and/or easier to detect
- **Maintainability -** the pipeline can be updated and extended more easily
- **Comparability and interoperability -** facilitates combining different datasets and building on top of existing systems
- **Data governance -** easier to maintain audit trails and implement data governance policies

Qualities of a good data pipeline

- Data standardisation
- Quality assurance
- Recurrent and reproducible processing
- Maintainability
- Performance and efficiency
- Security and data governance

Quality assurance (QA)

Data issues will happen - catch them early using QA checks, and take appropriate action.

Types of checks:

- **Syntactic checks** - check that data **structure** is correct
  - E.g. check for corrupted files, null fields, missing time range, duplicates
- **Semantic checks** - check data content is **interpretable** (i.e. “makes sense”)
  - E.g. check for impossibly-fast movements, devices with abnormal event volume, cell IDs with no known cell location

More details of QA checks in session 2.3.4

Quality assurance (QA)

Along with checks, record quality metrics (e.g. number of data issues, average/maximum values of fields)

- Routine monitoring of system health
- Some checks may require comparing to previous values (e.g. data volume outside normal range)
- Manual checks and debugging if problems are detected
- Later processing steps may use quality metrics, e.g. to filter out devices with low event count

Quality assurance (QA)

Detected quality issues should trigger actions:

- **Data cleaning** (drop invalid data)
- **Alert** relevant staff members if issues are detected
- Influence later **processing** stages, e.g.:
  - Skip production of outputs due to missing/incomplete data
  - Or produce outputs but with a data quality advisory
- Produce additional **diagnostic** outputs

QA checks should be **automated**, even for manually-triggered workflows.

Qualities of a good data pipeline

- Data standardisation
- Quality assurance
- Recurrent and reproducible processing
- Maintainability
- Performance and efficiency
- Security and data governance

Recurrent and reproducible processing

Minimise manual effort, and ensure outputs can be reproduced later if required.

- Process data with **reusable scripts** (using e.g. python, bash)
- **Automate** recurring tasks, e.g. using cron (simple time-based scheduling) or Airflow (more complex workflow orchestration)
- **Fault tolerance -** graceful error handling, automatic retries
- **Monitoring -** comprehensive logging and alerting
- **Data lineage** and **version control -** ensure we know not just that the pipeline is working, but also what was produced and how

Qualities of a good data pipeline

- Data standardisation
- Quality assurance
- Recurrent and reproducible processing
- Maintainability
- Performance and efficiency
- Security and data governance

Maintainability

- Clear, well-documented and version-controlled code
- Modular design - build pipeline from small independent tasks that can be separately debugged, updated and extended
- Components are tested before deployment
- Streamlined updates with minimal disruption
- Establish update process
  - Code review
  - Testing
  - Deployment/rollback process
  - Roles and responsibilities for maintenance

Qualities of a good data pipeline

- Data standardisation
- Quality assurance
- Recurrent and reproducible processing
- Maintainability
- Performance and efficiency
- Security and data governance

Performance and efficiency

Considerations for data pipeline performance:

- Data processing completes in a **timely manner**
- **Sufficient storage** (potentially for large time-series if the pipeline will be operating for a long time), including data backup
- Able to handle **growing data volumes** over time (planning ahead with over-spec hardware, and/or ensuring system can be scaled without redesigning architecture)
- **Optimisation**, to minimise unnecessary processing/storage costs

Qualities of a good data pipeline

- Data standardisation
- Quality assurance
- Recurrent and reproducible processing
- Maintainability
- Performance and efficiency
- Security and data governance

Security and Data Governance

Security considerations for a data pipeline:

- Access control
  - Restrict access to only those who need it, and record access permissions
- Server hardening and data encryption
  - Ensure nobody can gain access without authorisation
- Audit trails
  - Record who does what, so incidents can be caught and properly investigated

Security and Data Governance

Regulatory and organisational security compliance measures may include:

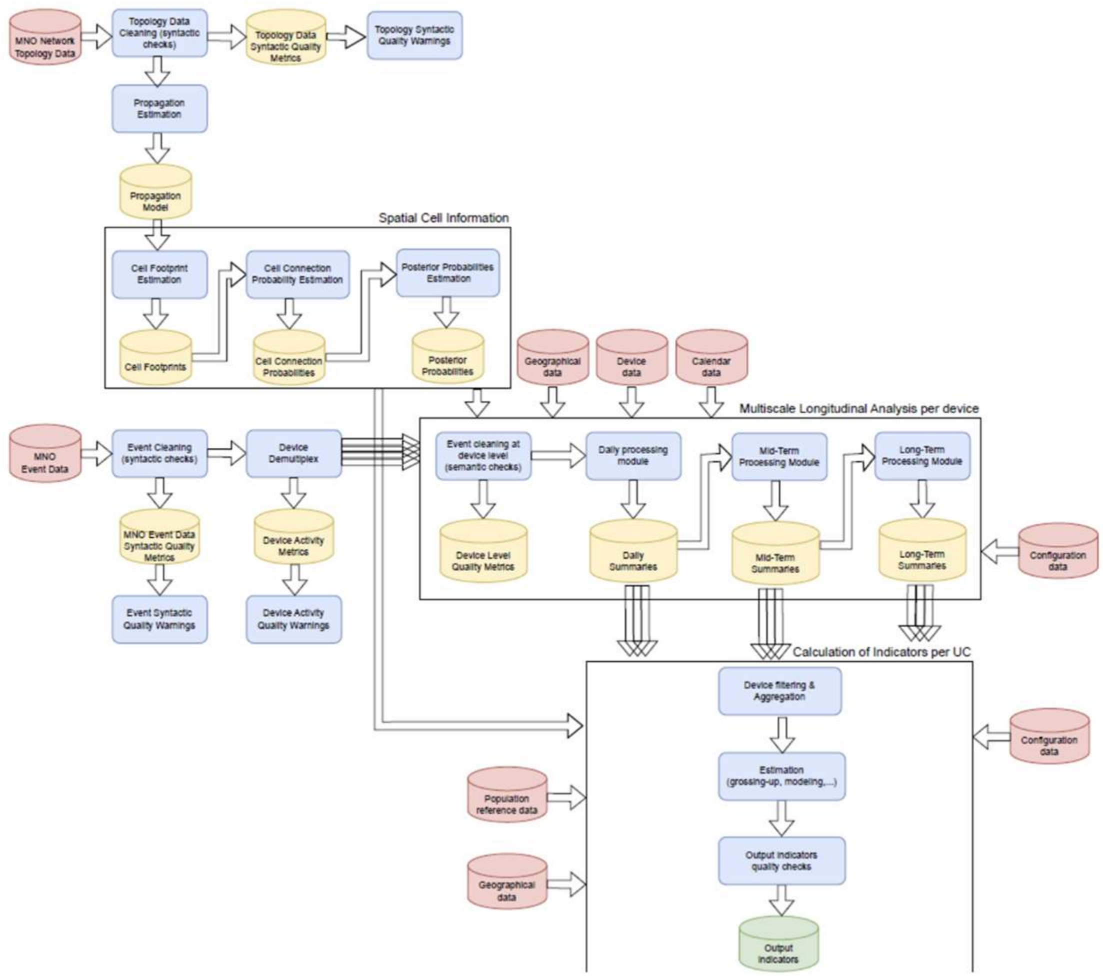
- **Data localisation** (e.g. keep personal data within premises/country of owner)
- **Incident response procedures** (clear process when something goes wrong)
- **Sensitivity labelling** (secret, confidential etc.; different handling rules for different sensitivity levels)
- **Data lineage** (e.g. to comply with individual consent withdrawal / objection)
- **Retention/deletion policies**
- **Privacy protection** (e.g. statistical disclosure control)
- More on this in session 2.3.3 (privacy by design)

Best practice for developing a pipeline

Stages of a data pipeline

Example: Eurostat Multi-MNO reference pipeline

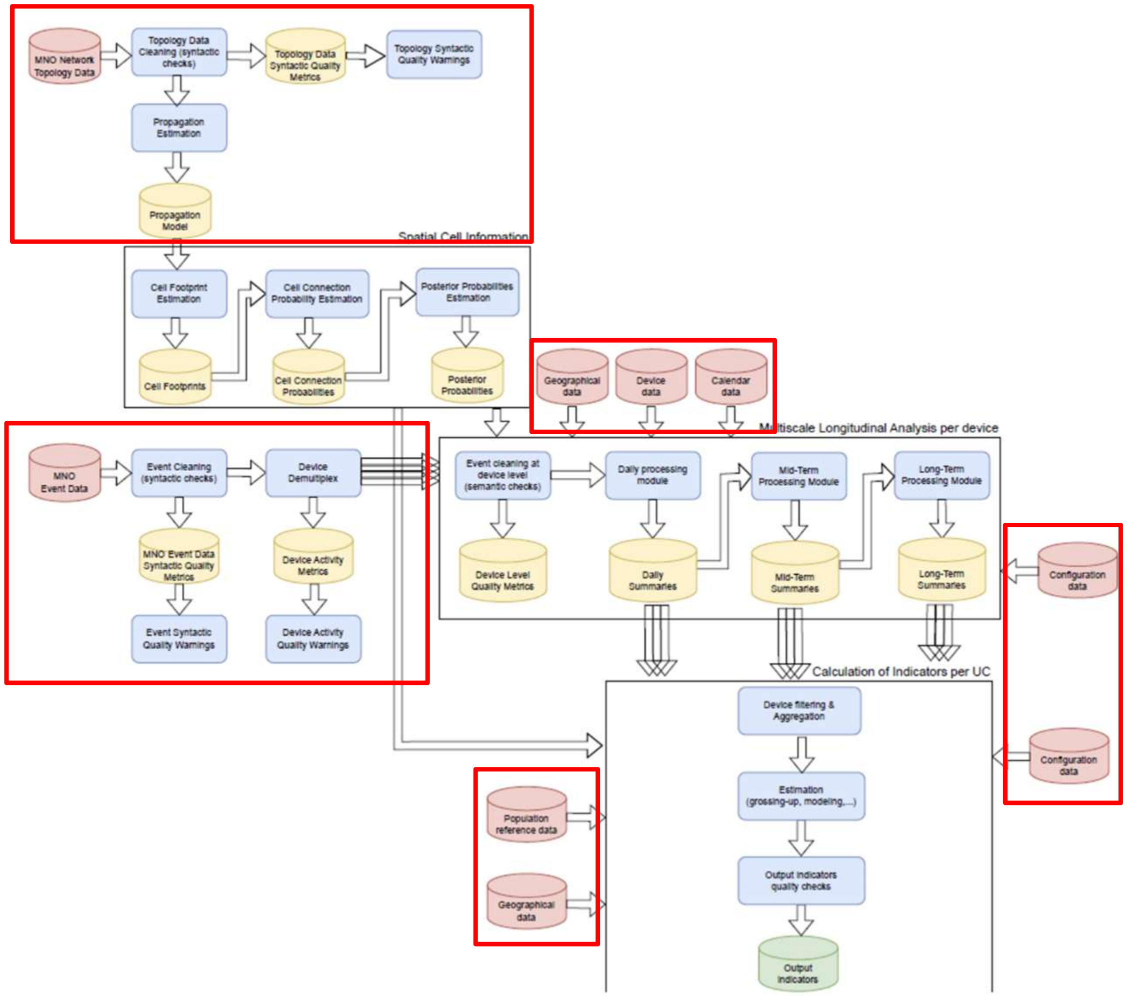
- Produces indicators at day/month/year time scales for official statistics
- Co-developed by a consortium (GOPA, Nommon, Positium, ISTAT, CBS)
- Tested on real MNO data in several European countries



GDF GLOBAL DATA FACILITY · GOBIERNO DE ESPAÑA · MINISTERIO DE ECONOMÍA, COMERCIO Y EMPRESA · THE WORLD BANK IBRD • IDA | WORLD BANK GROUP · ITU · FLOWMINDER.ORG · NOMMON · positium

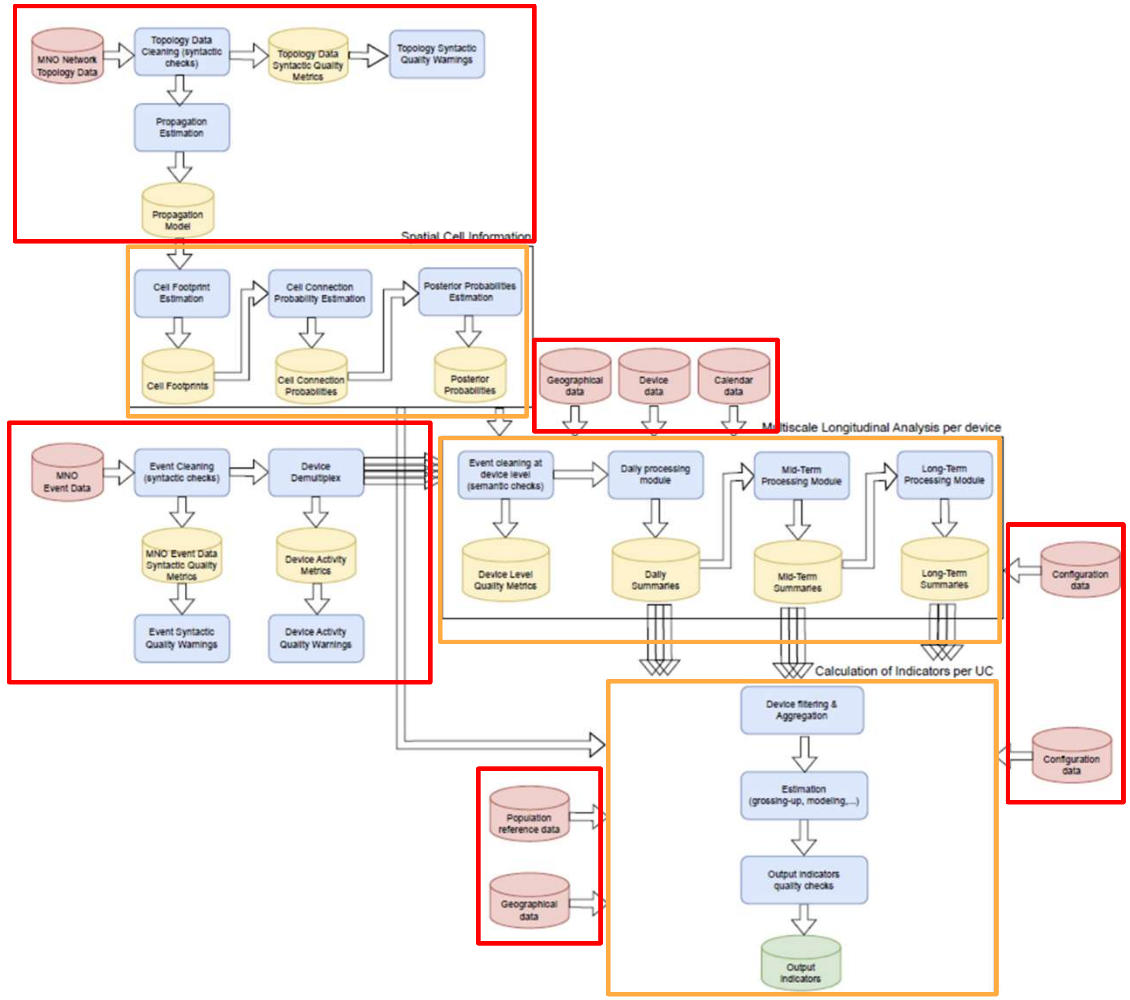
Stages of a data pipeline

1. Data ingestion



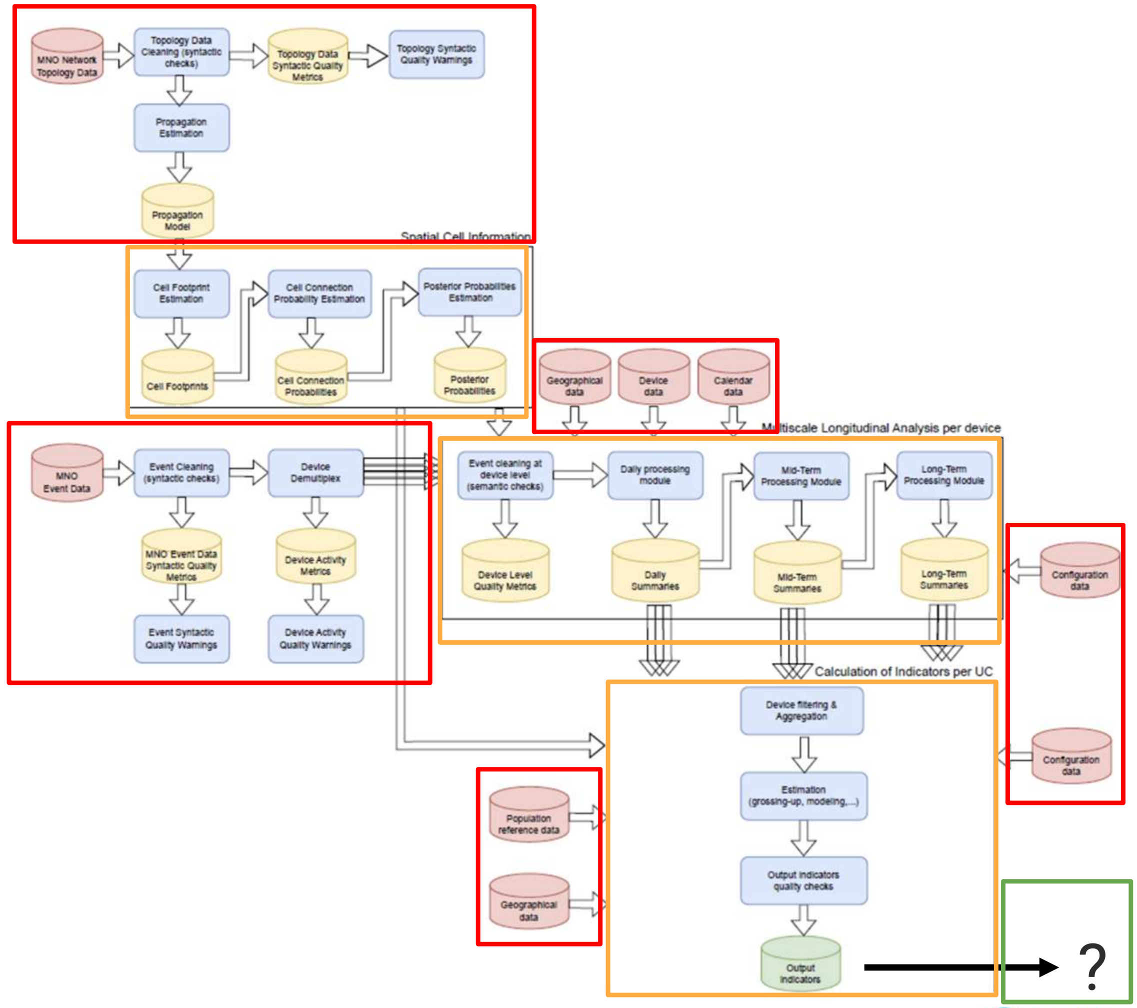
Stages of a data pipeline

1. Data ingestion
2. Data processing and production of outputs



Stages of a data pipeline

1. Data ingestion
2. Data processing and production of outputs
3. Dissemination

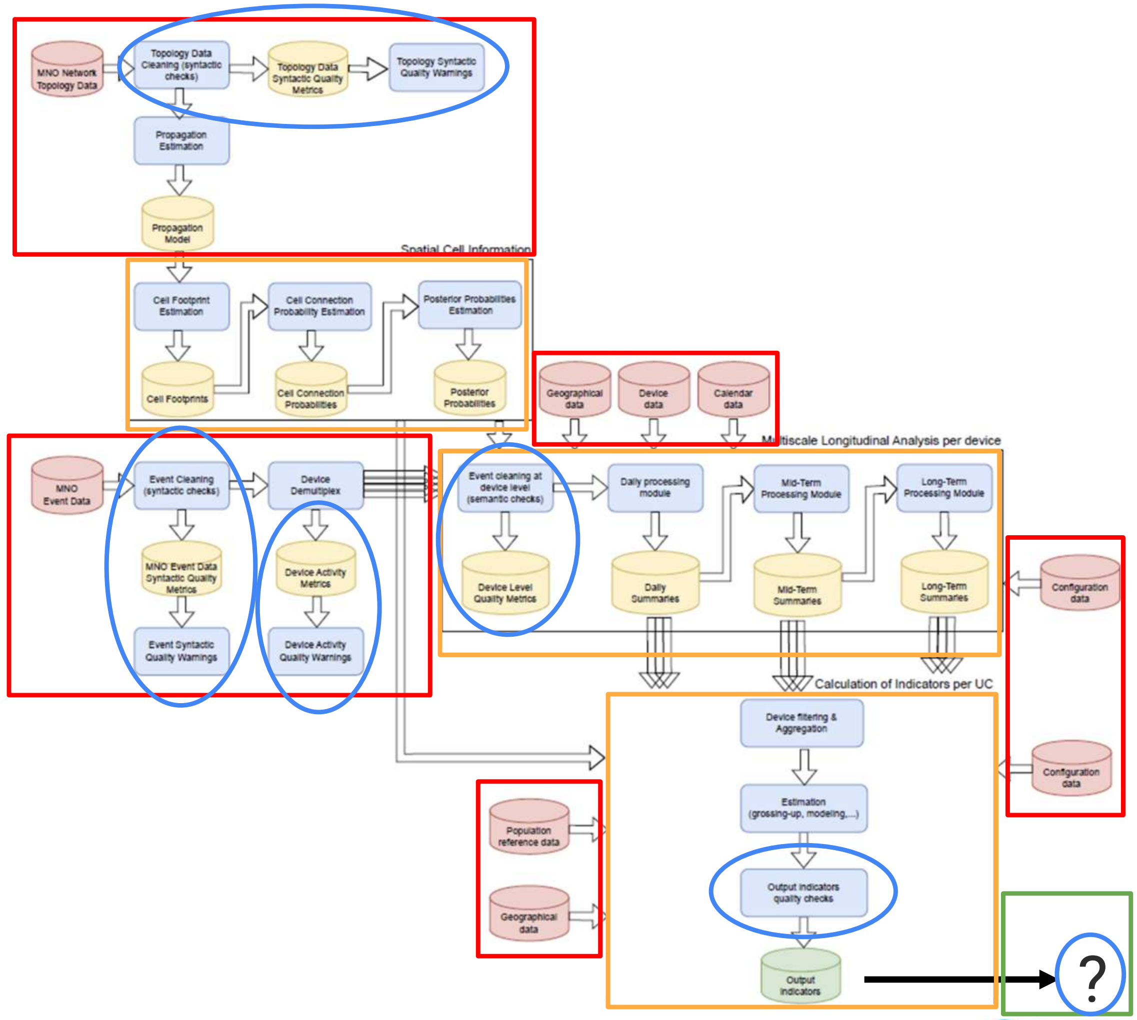


GLOBAL DATA FACILITY THE WORLD BANK FLOWMINDER.ORG NOMMON positium

Stages of a data pipeline

1. Data ingestion
2. Data processing and production of outputs
3. Dissemination

Throughout: QA and alerting



GDF GLOBAL DATA FACILITY · GOBIERNO DE ESPAÑA MINISTERIO DE ECONOMÍA, COMERCIO Y EMPRESA · THE WORLD BANK IBRD • IDA | WORLD BANK GROUP · ITU · FLOWMINDER.ORG · NOMMON · positium

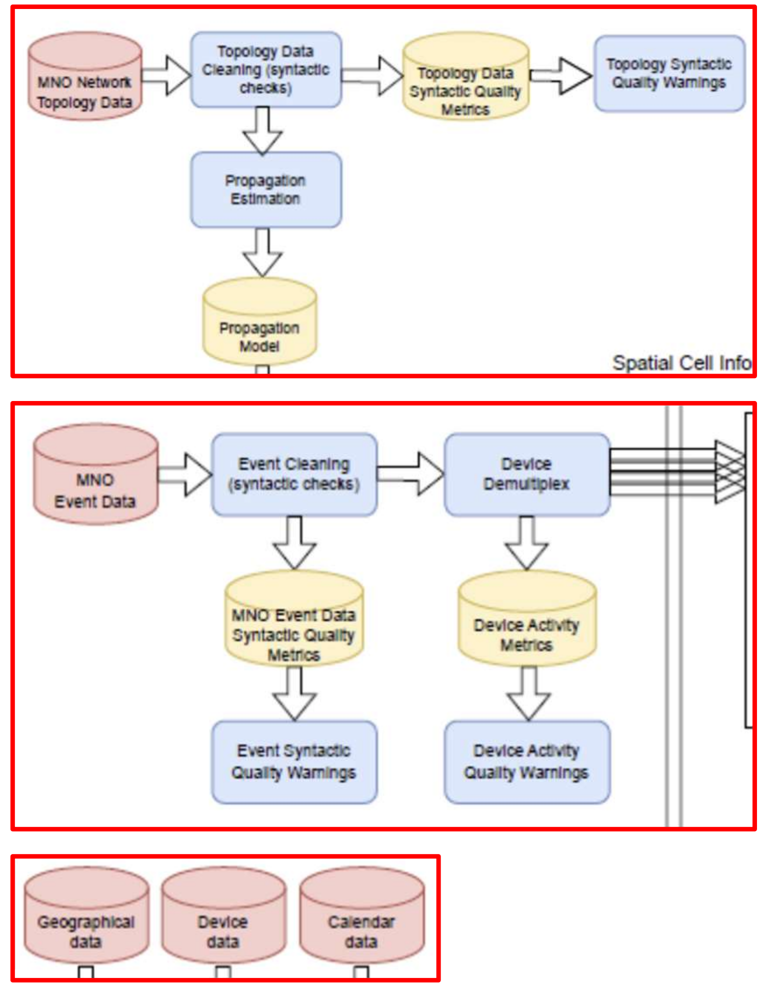
Data ingestion

- Extract, Transform, Load (ETL) - or ELT
- Input data **standardisation** - transform data from potentially disparate sources and structures into a known structure on which later processing stages can operate
- Ensure data **integrity** - catch problems early before they propagate through the pipeline
    - Syntactically invalid data must either be removed, or raise an error to require manual intervention
- Frequency of ingestion may be influenced by data source - not necessarily the same as frequency of production of outputs
    - Different datasets may be updated at different frequencies, and at different points in the system (possibly into different databases)

Data ingestion

Eurostat example:

- **Network topology data** (coverage maps and/or cell parameters) extracted, cleaned, and transformed into a spatial propagation model per cell - ideally daily
- **MNO event data** extracted, cleaned, and transformed into an ordered time series of events per device - ideally daily
- **Other auxiliary datasets** (geography, population etc.) may be updated less frequently, and ingested at different stages (e.g. population data only required for the final stage of processing)



Data processing and production of outputs

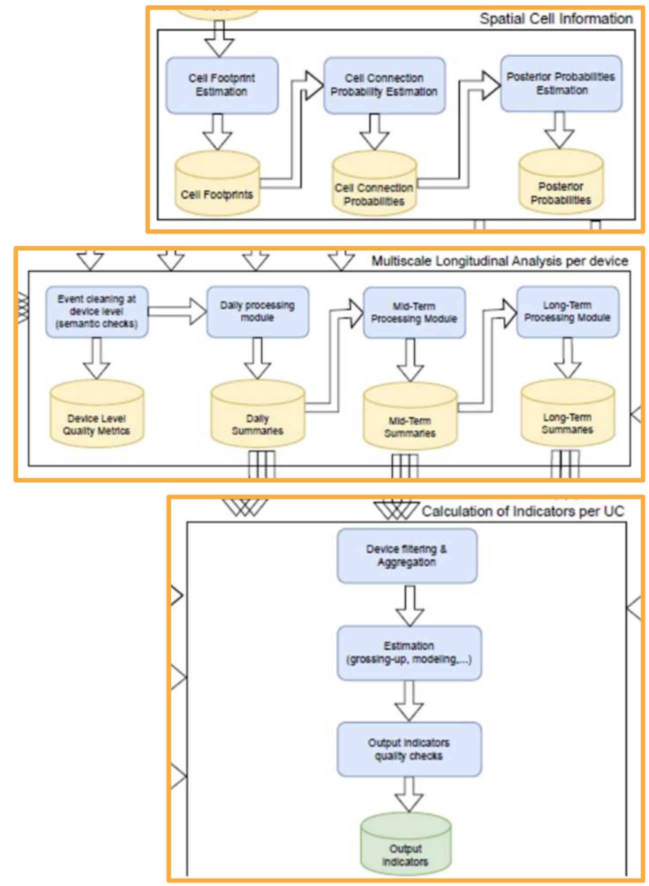
- Chain of processing steps - each takes one or more data objects as input and creates/updates other data objects
- First processing steps operate on the cleaned source data from the ingestion stage
- Final processing steps produce the outputs
  - Indicators
  - Diagnostic data
  - And/or visualisations/reports/summaries
- Can be any number of steps in between, producing intermediate data objects

Details of processing/analysis methods in section 2.4

Data processing and production of outputs

Eurostat example:

- **Multiple processing stages**
  - Transform cell coverage information to location probabilities
  - Per-device analysis (daily, mid-term, long-term)
  - Aggregation and calculation of indicators
- **Modular design** - independent functions that can be replaced or added to support different use cases
- Each function can **access all data objects** produced by any previous function
- But later functions cannot update earlier data objects (**directed** flow of information)



Dissemination

Will depend on the use case and end-users, e.g.:

- Push data to a repository (public or private)
- Update interactive dashboard
- Send PDF reports by email

Several factors influence the type, format and frequency of outputs, such as:

- **Purpose** for which outputs are used
- Technical/data **literacy** of end-users
- **Environment** in which outputs will be accessed

Eurostat example: out of scope (pipeline produces indicators, but does not specify where/how these should be conveyed to users)

QA and alerting

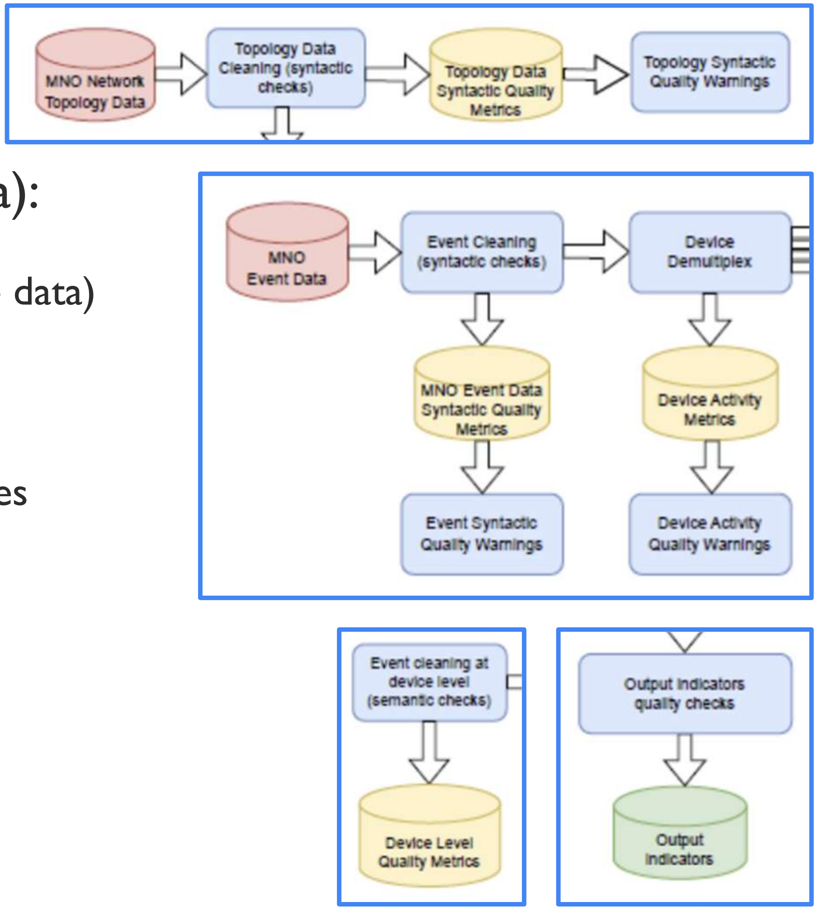
- Integrated **throughout** the pipeline - quality checks at every stage
- Check **validity** and **reliability** of data
- Produce and record **diagnostic metrics** for monitoring and review
  - E.g. number of invalid rows; number of active subscribers
- Take **action**, e.g.:
  - Exclude invalid/problematic data
  - Halt production of some or all outputs
  - Add quality warnings to outputs
  - Some quota of problems is to be expected - set error threshold
- **Alert** responsible person(s) if issues are detected
  - E.g. send automated email with summary of the problem
- May involve **manual review** stages (human-in-the-loop)
  - E.g. review report before release to assess impact of data issues

More details in session 2.3.4

QA and alerting

Eurostat example:

- Syntactic checks (network topology & event data):
  - Purge malformed data during ingestion
  - Produce quality metrics (information about issues in the data)
  - Launch warnings if issues are detected
- Device activity metrics
  - E.g. number of events per hour
  - Can be used later to filter out devices for some use cases
  - Launch warnings if issues are detected
- Device-level semantic checks
  - E.g. impossibly fast movements
  - Remove events that "don't make sense"
- Quality checks on output indicators
  - Produce quality indicators in line with official statistics



Where should processing take place?

Pipeline may be split between multiple environments (e.g. on-premises and cloud infrastructure; data owner's and service provider's systems). Several factors:

- Data **localisation** policies/regulations
  - Personal or commercially-sensitive data may need to remain within MNO
  - Anonymised, aggregated data can be shared more freely
- **Integration** with existing systems/datasets
  - E.g. bias adjustment using reference data belonging to NSI
- Data **size**
  - Earlier-stage processing of individual-level data likely to require larger compute resources
  - Later stages on smaller aggregated datasets are cheaper
- Available **hardware**
  - MNO will have infrastructure for processing large MPD datasets
- Intellectual "**ownership**"
  - Different partners may want/need control of different processing stages (e.g. technical service provider specifying methods/models)

Where should processing take place?

Eurostat example: combines data from multiple MNOs. Allows for two possible scenarios:

Basic approach

1. All individual-level data processed within MNO premises
2. Aggregated outputs transferred to NSI for adjustment and modelling

Advanced approach

1. Raw MPD data processed within MNO premises
2. Individual-level daily summaries transferred to secure enclave for combined processing
3. Aggregated outputs transferred to NSI for adjustment and modelling

Where should processing take place?

Eurostat example: combines data from multiple MNOs.

Common between the two scenarios:

- Raw MPD processed within MNO (where infrastructure already exists)
- Data minimisation - only necessary data transferred out of data owner's premises
- Final modelling/estimation handled within NSI, which has method ownership and necessary auxiliary datasets

Challenges of large data size

- Data processing capability (parallelisation, memory, disk read/write speed)
  - Impacts how **quickly** outputs can be produced
- Storage capacity
  - Impacts how long data can be **retained** (for archiving, re-calculating outputs, and regulatory compliance)
- Network bandwidth and latency
  - For **dissemination** of outputs, and/or internal **data transfer** within the pipeline
- For optimisation, identify **bottlenecks** (CPU, memory, disk I/O, network data transfer; time spent in each processing stage)
  - Don't just guess which bit is slow, or you may end up spending a lot of effort with little impact

Summary

Summary

- MNO data for pipeline input may come from multiple sources managed by different teams
- Good data pipelines should be automated, maintainable, efficient and secure, with QA, monitoring, logging and alerting integrated throughout
  - See session 2.3.3 for more on data security
- Pipeline design is influenced by factors such as end user requirements, data size, available hardware, data/method ownership and applicable regulations.

Further reading

Case studies and further reading

Eurostat Multi-MNO project: https://cros.ec.europa.eu/landing-page/multi-mno-project

- Methodological framework: https://cros.ec.europa.eu/book-page/methodology-framework-high-level-architecture-requirements-use-cases-and-methods

FlowKit (Flowminder's MPD analysis platform): https://flowkit.xyz

- White paper: https://www.flowminder.org/resources/publications-reports/flowkit-unlocking-the-power-of-mobile-data-for-humanitarian-and-development-purposes

Handbook on MPD for Official Statistics - UN Global Working Group on Big Data for Official Statistics (2019):
https://unstats.un.org/bigdata/task-teams/mobile-phone/MPD%20Handbook%2020191004.pdf

The GDF-MPD window is a multi-partner window funded by the Spanish Government, Ministry of Economy, Commerce and Business. The GDF-MPD program is jointly managed by DEC (DECDG, DIME), Poverty and Digital Development Global Practices in technical partnership with the International Telecommunications Union (ITU).